

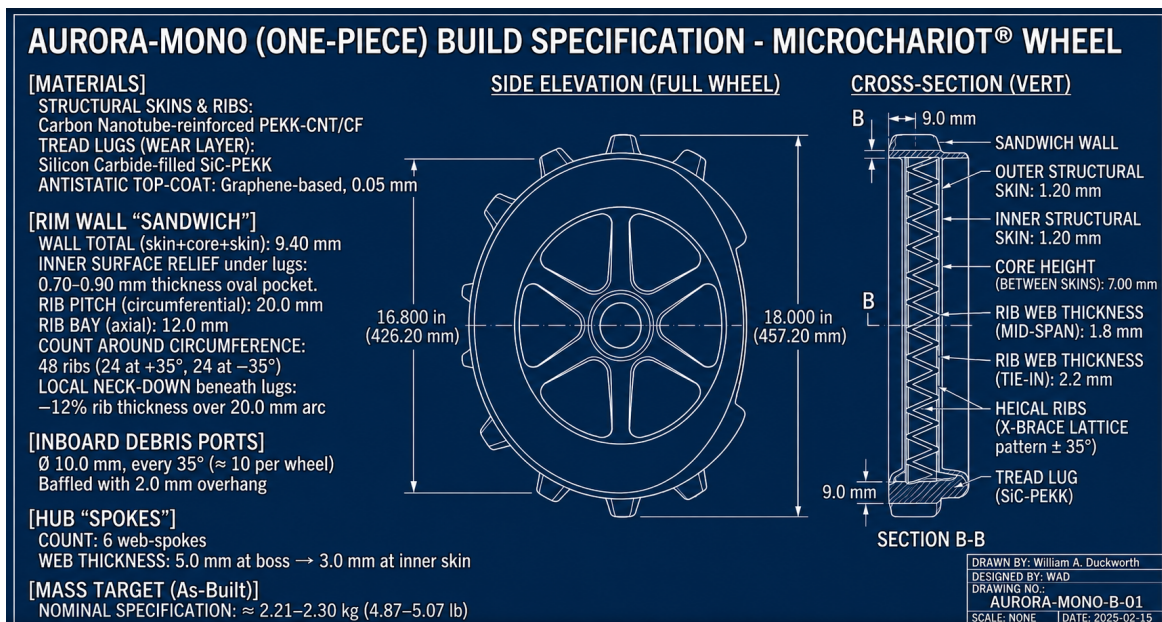
AURORA-Mono

(One-Piece) Rover Wheel

Design Specification & Screening-Level Validation

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Revision 2.0 · 2026



Build specification drawing AURORA-MONO-B-02. Side elevation, vertical cross-section B–B showing the sandwich wall with the ±35° X-brace helical rib lattice, and the full materials / rim-wall / spoke / mass-target callouts.

Abstract

AURORA-Mono is a one-piece composite rover wheel engineered to the NASA MicroChariot interface envelope. This document specifies Revision 2.0 of the design, which integrates four targeted improvements identified by a closed-form Python validation campaign: a 1.0 mm unfilled-PEKK compliant interlayer between the SiC-PEKK tread and the PEKK-CNT/CF outer skin; a reformulated SiC-PEKK tread with reduced coefficient of thermal expansion ($\alpha_{\text{tread}} = 20 \text{ ppm/K}$); chamfered lug-base edge geometry; and Ti-6Al-4V hub bolts torqued to 50% of proof load. With these

changes the wheel meets a safety factor ≥ 2.0 on every analytically resolved failure mode and ≥ 1.5 on the static-peel mode at the lug-skin bond under lunar diurnal thermal cycling. The complete validation campaign (13 reproducible Python screening models) is documented in the companion *aurora-mono-simulations* folder of the repository.

Revision history

Rev	Date	Notes
1.0	2026-02-15	Initial design specification.
2.0	2026-05-22	Updated baseline integrating four design changes from screening validation campaign. New valid

1. Envelope & Interfaces (MicroChariot Compliant)

- Overall OD at lug tips: **18.000 in** (457.20 mm)
- Overall width: **8.000 in** (203.20 mm)
- Rim OD (outer structural skin, pre-tread): **16.800 in** (426.72 mm)
- Hub pilot: **Ø 3.185 in** (80.899 mm)
- Bolt circle: **Ø 4.000 in** (101.60 mm), 8× #10-32 UNF THRU, equally spaced (45°)
- Jack bolts: 2× #10-32 UNF THRU on the same Ø 4.000 in circle, 180° apart
- Pin holes: 2× Ø 0.2502 ±0.001 in (6.356 ±0.025 mm) on Ø 4.000 in circle, 180° apart
- Hub adapter plate thickness (composite): 6.0 mm with 12.0 mm diameter pad bosses at holes
- Keep-out zone: honors 4.000 in (W) × 1.750 in (H) box from NASA drawing; no intrusion

1.1 Tolerances (unless noted)

- Hub pattern features: ±0.05 mm (±0.002 in)
- Critical radii (OD, width): ±0.25 mm (±0.010 in)
- All others: ±0.5 mm (±0.020 in)

2. Materials

Rev 2.0 introduces a compliant interlayer and a reformulated tread to address the thermal-cycling static-peel concern identified by the validation campaign (see Section 11). The full material stack from outer surface to hub:

- **Tread lugs (wear layer):** Reformulated Silicon-Carbide-filled PEKK (SiC-PEKK), ~50 vol% SiC. $\rho \approx 1.80$ g/cc; $\alpha_{\text{tread}} \approx 20$ ppm/K (reduced from 29 ppm/K in Rev 1.0).
- **Compliant interlayer (NEW in Rev 2.0):** Unfilled PEKK shim, 1.0 mm thick, between the SiC-PEKK tread and the PEKK-CNT/CF outer skin. Bonds chemically to both same-family PEKK surfaces. $\rho \approx 1.30$ g/cc; $E \approx 4$ GPa; $\alpha \approx 45$ ppm/K.
- **Outer/inner structural skins & helical ribs:** Carbon Nanotube-reinforced, Carbon Fiber-reinforced Polyetherketoneketone (PEKK-CNT/CF). $\rho \approx 1.45$ g/cc; $E \approx 50$ GPa (in-plane); $\alpha \approx 10$ ppm/K (in-plane).
- **Hub bolts (UPDATED in Rev 2.0):** Ti-6Al-4V #10-32 UNF (replaces A286 stainless in Rev 1.0). Torqued to **50%** of proof load.
- Hub hole sleeves (optional): Ti-6Al-4V, wall 0.5 mm (press-fit) for metal interfaces.
- Antistatic top-coat (non-lug areas): graphene-based, 0.05 mm (vacuum-compatible).

3. Rim Wall “Sandwich” Cross-Section

All dimensions radial from wheel center.

- **Tread lug (SiC-PEKK):** 9.0 mm nominal lug height + 6.24 mm carcass base (see Section 6).
- **Compliant interlayer (NEW):** 1.0 mm unfilled PEKK between the tread carcass and the outer structural skin.

- **Outer structural skin:** 1.20 mm thick, continuous and sealed.
- **Core height between skins:** 7.00 mm nominal.
- **Inner structural skin (toward hub):** 1.20 mm thick, continuous.
- **Wall total (interlayer + skin + core + skin): 10.40 mm** (Rev 1.0 was 9.40 mm; +1 mm for interlayer).
- Inner surface local relief under lugs (flex zone): reduce inner-skin-side wall to 0.70–0.90 mm within a 28 × 20 mm oval directly beneath each lug, machined/pocketed from inside.

3.1 Helical ribs (between skins)

- Pattern: alternating $\pm 35^\circ$ helix families (X-brace lattice). Unchanged from Rev 1.0.
- Rib web thickness: **1.8 mm** at mid-span, **2.2 mm** at skin tie-in.
- Rib pitch (circumferential spacing of nodes): 20.0 mm between successive ties on each skin.
- Rib bay (axial spacing across width): 12.0 mm, yielding ≈ 17 bays across the 203.20 mm width.
- Count around circumference: **48 ribs total** (24 at $+35^\circ$, 24 at -35°), evenly distributed.
- Local neck-down beneath lug footprints: -12% rib thickness over a 20.0 mm arc length (adds micro-compliance).

3.2 Inboard debris ports (NOT on tread face)

- $\varnothing$ 10.0 mm, one every 35° (≈ 10 ports per wheel).
- Lip-baffled with a 2.0 mm deep overhang; no line-of-sight to the outer skin.

4. Hub “Spokes” (Torque Web, Inside Cavity)

Thin composite webs run from the hub adapter to the inner rim skin and are not visible in the tread wall. Unchanged from Rev 1.0.

- Count: **6 web-spokes**, equally spaced.
- Web thickness: 5.0 mm at hub boss tapering to 3.0 mm at rim inner skin; aero-tri profile with 5.0 mm fillets both ends.
- Lightening pockets: triangular cutouts leaving ≥ 30.0 mm minimum shear path; web area open ratio $\approx 45\%$.
- Purpose: torque transfer and stiffness with minimal mass.

5. Tread System

Integral tread, molded onto the sealed outer skin through the 1.0 mm compliant PEKK interlayer. Base (carcass) thickness to reach the 18.000 in OD with 9.0 mm lugs is $t_{base} = \mathbf{6.24\text{ mm}}$ (from outer skin up to the lug root). The construction places a thin outer shell (~ 1.0 mm) over the lattice peaks; the remaining ~ 5.2 mm is mostly rib crests and void, yielding a 10–12% solid equivalent.

5.1 Lug geometry (two staggered rows, dirt-bike style)

- Height: 9.0 mm nominal; alternating 8.0 mm / 10.0 mm every other lug within a row.
- Plan shape: chevron blocks; rake angle 60° leading face, trailing chamfer 22° .

- Block length (circumferential): 45.0 mm long / 35.0 mm short, alternating (± 5.0 mm).
- Block width (axial): 14.0 mm; row centers at ± 30.0 mm from mid-plane.
- Row phase shift (helical wrap): $\frac{1}{2}$ -pitch (≈ 20.0 mm) between rows.
- Shoulder side-biters: every 3rd lug gets a 3.0 mm tall, 4.0 mm long buttress at the outer shoulder.
- Void ratio (contact patch): 75–80% (coverage 19–20%).
- Root fillets: 1.2 mm on all interior edges to avoid chipping.
- External corner radii: 0.6–0.8 mm.
- **Chamfered lug-base edge geometry (NEW in Rev 2.0):** 1.5 mm $\times$ 45° chamfer at the perimeter of each lug footprint to spread the stress singularity at the lug-skin bond edge under thermal-cycle and cresting loads.

5.2 Lug-to-skin bond stack (UPDATED in Rev 2.0)

- **Layer 1 (top):** SiC-PEKK tread lug.
- **Layer 2 (NEW):** 1.0 mm unfilled-PEKK compliant interlayer.
- **Layer 3:** PEKK-CNT/CF outer structural skin.
- Bonded via three-stage same-family chemical bond: tread / interlayer co-mold at 355–365 °C, 0.6–0.8 MPa, dwell ≥ 20 min; interlayer / skin bond same process window.
- Anti-peel keys (retained from Rev 1.0): 3 concentric 0.6 mm deep $\times$ 1.8 mm wide shallow grooves in the outer skin surface under each lug; fill with interlayer PEKK during molding.

6. Hub Bolt Joint

Eight #10-32 UNF mounting bolts plus 2 jack bolts and 2 alignment pins on the $\varnothing 4.000$ in bolt circle. The bolt material and preload spec were updated in Rev 2.0 to recover SF ≥ 2.0 on both bolt yield and pad-boss compression; geometry is unchanged from Rev 1.0.

- **Bolt material (UPDATED):** Ti-6Al-4V (was A286 stainless in Rev 1.0). Standard aerospace fastener spec; $\sim 40\%$ lighter than A286.
- **Preload spec (UPDATED):** torque to **50% of proof load** (was 70% in Rev 1.0). Tighten with a calibrated torque wrench or use load-indicating washers.
- All other joint geometry (bolt size, bolt circle, pad-boss diameter, alignment pins, jack bolts) unchanged from Rev 1.0.

7. Manufacturing Sequence (UPDATED)

Same overall flow as Rev 1.0, with one additional step (4) inserting the compliant interlayer between skin consolidation and tread compression molding.

- 1 Print lattice core and hub web in PEKK-CNT/CF using high-temperature additive manufacturing.
- 2 Lay inner/outer skins (PEKK-CNT/CF tape) and co-consolidate in an autoclave (≈ 360 °C, 0.7 MPa, 2–3 hours).
- 3 Machine the hub pattern, debris ports, and inner-surface relief pockets under the lug locations.

- 4 **(NEW)** Lay the 1.0 mm unfilled-PEKK compliant interlayer on the prepared outer skin; bond at 355–365 °C, 0.6–0.8 MPa, dwell ≥20 min.
- 5 Compression-mold the reformulated SiC-PEKK tread onto the interlayer using the chevron cavity tool with chamfered lug-base detail (engaging the anti-peel keys).
- 6 Apply the antistatic top-coat, avoiding lug faces.
- 7 Install Ti-6Al-4V hub bolts to 50% proof load using a calibrated torque wrench.
- 8 Final QC: dimensional validation, mass validation (target 2.30–2.40 kg), and non-destructive inspection (ultrasound / CT).

8. Critical Dimensions Summary

Feature	Dimension
OD at lug tips	18.000 in (457.20 mm)
Width	8.000 in (203.20 mm)
Rim OD (pre-tread)	16.800 in (426.72 mm)
Outer / inner skin	1.20 mm each
Compliant interlayer (Rev 2.0)	1.00 mm unfilled PEKK
Core height	7.00 mm
Wall total	10.40 mm
Helical rib web	1.8 mm (mid-span), 2.2 mm (tie-in)
Helix angle	±35°
Ribs (count)	48 total (24 positive / 24 negative)
Hub web spokes	6 spokes, 3.0–5.0 mm thick
Tread base (carcass)	6.24 mm (10–12% solid equivalent)
Tread formulation (Rev 2.0)	SiC-PEKK, ~50 vol% SiC
Lug height	9.0 mm nominal (8.0 / 10.0 mm alternating)
Lug width / length (pitch)	14.0 mm / 35.0 / 45.0 mm alternating
Lug-base chamfer (Rev 2.0)	1.5 mm × 45°
Row offset	±30.0 mm from mid-plane
Void ratio	75–80% (coverage 19–20%)
Anti-peel keys	3 grooves, 0.6 mm × 1.8 mm
Debris ports (inboard)	Ø 10.0 mm, every 35°, baffled
Hub bolts (Rev 2.0)	8× Ti-6Al-4V #10-32 UNF, 50% proof preload

Table 1. Critical dimensions summary, Rev 2.0.

9. Mass Target (As-Built)

- **Rev 2.0 specification:** 2.30–2.40 kg (5.07–5.29 lb). Rev 1.0 was 2.21–2.30 kg; the 1.0 mm unfilled-PEKK interlayer adds ≈60–90 g, partially offset by the lighter Ti-6Al-4V bolts (≈5–10 g savings across 10 fasteners).
- Mass optimization margin: reduce total surface coverage down to 19%, or implement hollow-web configurations within the lugs, saving 80–120 g while maintaining the identical external footprint.

10. Validation Campaign

Rev 2.0 was shaped by a 13-check Python validation campaign that evaluated every identifiable failure mode at screening fidelity. All scripts are reproducible (fixed random seed), with CSV outputs and plots, and live in the companion *aurora-mono-simulations/* folder of the repository. Headline results:

Check	Mode	Headline result (Rev 2.0)
Strip stress	Skin tensile under impact	Min SF 3.14, 0 fracture flags
Sensitivity sweep	±50% on dominant params	Min SF >2.1 in all sweeps
Lug shear	Bond shear under traction	SF 13–27
Lug peel (driving)	Bond peel from eccentric load	SF 6.1, ≥1.8 across sweep
Fatigue (driving)	Miner's-rule over 696k cycles	D ≈ 0, life >7,000 km
Helical rib lattice	Per-rib stress, buckling, shear	SF >3 worst-case
Thermal cycling	Static peel from diurnal swing	Baseline 0.88 → Rev 2.0 SF 4.85
Viscoelastic relax.	Prony + TTS, not FEM	Confirms static, not fatigue, mode
Hub bolt joint	Preload, shear, pad compression	Baseline 1.38/1.82 → Rev 2.0 2.14/3.33
Launch load	QS 6g + Miles random vib	Driving loads govern, not launch
Static-peel iteration	Recommended Rev 2.0 mitigations	Stack identified, SF 4.85
Bolt-joint iteration	Recommended Rev 2.0 changes	Stack identified, both SFs > 2
Modal analysis	Closed-form 6-mode estimate	3 modes in 50–800 Hz band; FEM needed

Table 2. Validation campaign summary. All checks run against the Rev 2.0 design. See *aurora-mono-simulations/README.md* for the full method, sensitivity sweeps, and limitations of each check.

10.1 Selected validation plots

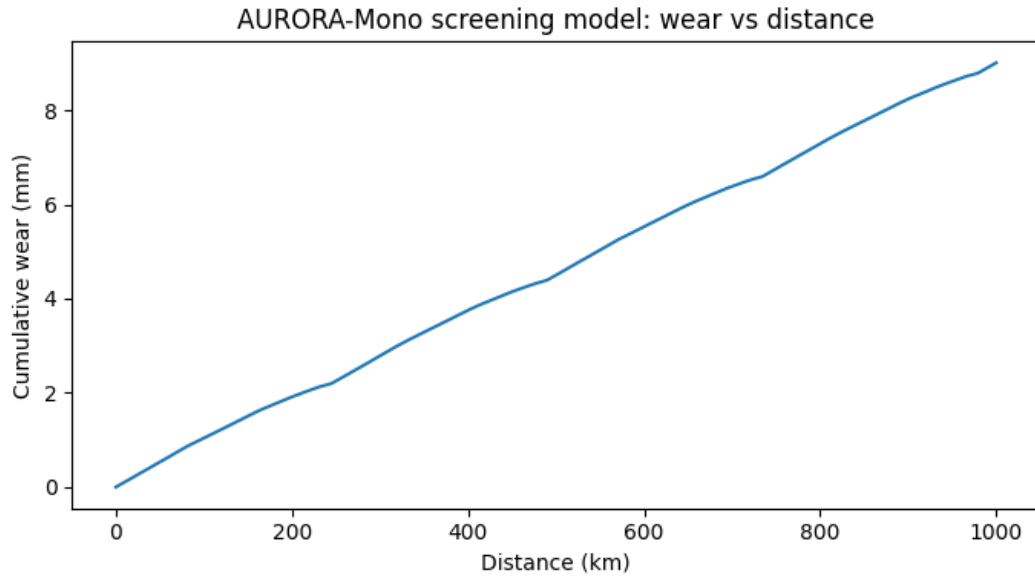


Figure 1. Cumulative tread wear vs distance over 1000 km of stochastic lunar driving (main screening model). Final wear 9.0 mm against 9.0 mm nominal lug height.

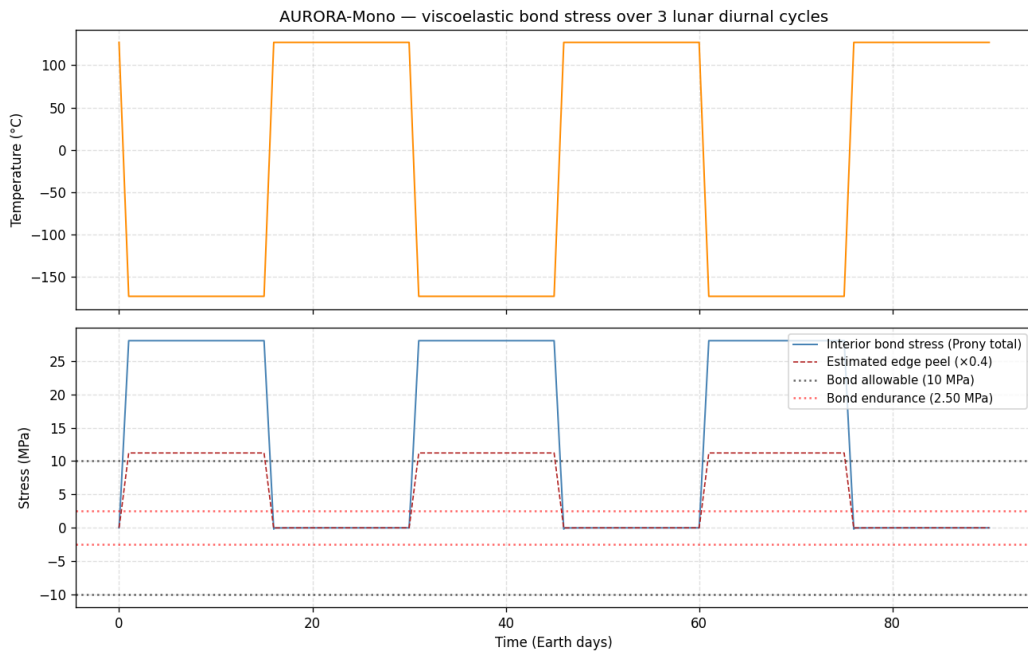


Figure 2. Bond stress over three lunar diurnal cycles in the Rev 1.0 design (no interlayer). Peak peel ~11 MPa exceeds the bond's 10 MPa static allowable on first cool-down. Rev 2.0 incorporates the 1.0 mm compliant interlayer (plus reformulated tread and chamfered lug edges) that drops peak peel to ~2 MPa, SF 4.85.

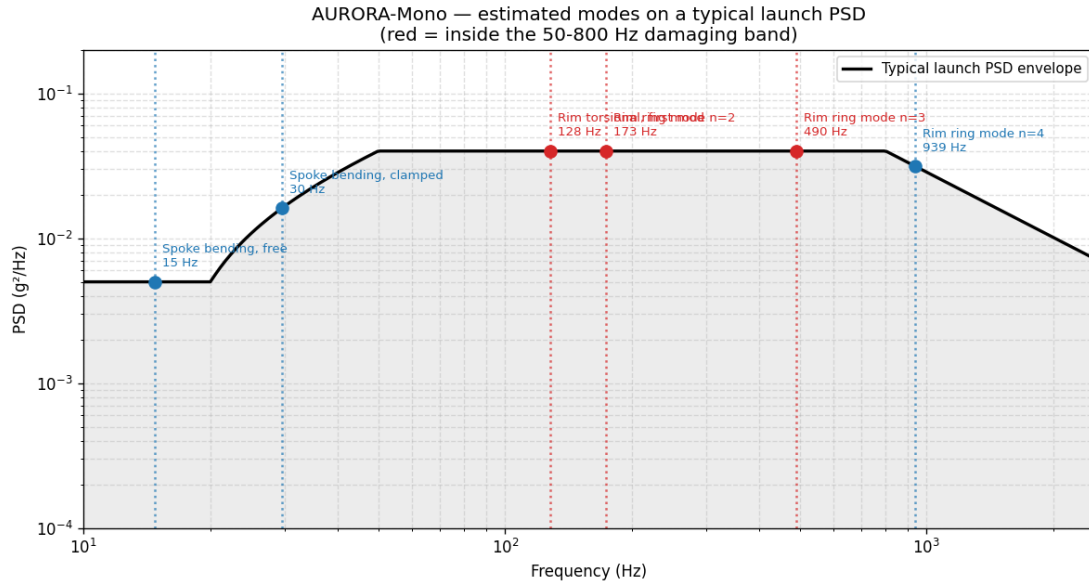


Figure 3. Estimated wheel natural frequencies (closed-form, not 3D eigenanalysis) overlaid on a typical launch random-vib PSD envelope. Rim torsional, $n=2$, and $n=3$ modes sit inside the 50–800 Hz damaging band (red). Coupled-loads FEM is needed to convict or acquit launch survivability at this fidelity.

11. Open Work & Next-Fidelity Steps

The screening campaign documented in Section 10 is the analytical envelope of what closed-form Python models can resolve for this design. The next-fidelity items below all require either real solver tooling, coupon test data, or physical hardware:

11.1 FEM / coupled-loads

- **3D viscoelastic FEM** of the lug-skin region under the diurnal cycle with measured Prony coefficients — validates the Rev 2.0 static-peel SF 4.85 from screening.
- **3D random-vibration FEM or coupled-loads analysis with the rover suspension** — three wheel modes sit inside the launch peak band per Section 10; the Miles' SDOF approximation under-estimates cumulative response.
- **3D truss/solid FEM of the rib lattice** — the lattice check treats ribs as pin-ended struts; the real critical location is the rib-to-skin joint.
- **Fracture-mechanics peel analysis** using G_c (Paris-law crack growth) for the actual co-mold bond, replacing the stress-based peel screening.
- **Bolt-joint creep + fatigue under thermal cycling** — the bolt screening is initial-condition only; PEKK pad-boss creep at +127 °C would relax preload over a long mission.

11.2 Coupon tests

- CTE values for SiC-PEKK and PEKK-CNT/CF in the actual Rev 2.0 production formulations.
- Co-mold bond strength (shear, peel, G_c , S-N) for the new three-layer stack (SiC-PEKK / unfilled-PEKK / PEKK-CNT/CF).
- Wear coefficient for the reformulated SiC-PEKK against JSC-1A lunar regolith simulant.

- Prony-series relaxation coefficients for the SiC-PEKK and unfilled-PEKK at +127 °C.

11.3 Operational scope not yet checked

- Steering / scrub loads if the rover steers.
- Radiation environment (UV, GCR, SPE) degradation on PEKK over the mission life.
- A real thermal model (radiation balance + 1D conduction through the sandwich wall + regolith contact + sun/shadow cycling), replacing the prescribed temperature schedule used in the screening model.

11.4 Hardware

Physical prototype build and bench test is out of scope for this study. Fabrication cost for a single qualified composite wheel of this geometry (PEKK feedstock, autoclave time, custom chevron compression mold, NDI) is estimated in the \$15K–\$50K range — not achievable on a hobby budget, and not the goal of a screening campaign. The next physical step would be coupon-level material qualification, not a full wheel.